Listening to the Ocean: Introduction to Underwater Sound

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Learning objectives

By the end of this unit, you should be able to:

- Explain what underwater sound is.
- Define frequency, amplitude, period and wavelength.
- Distinguish biological, anthropogenic and geophysical sound sources.
- Recognise waveform and spectrogram representations of underwater sound.

What is hydroacoustics?

Hydroacoustics is the study of the generation, propagation and detection of sound in water.

Hydrophones detect underwater pressure fluctuations and convert them into electrical signals. Recording systems can then digitise and store these signals for analysis.

Hydroacoustics is used to study:

- marine mammals and other biological sounds
- vessel noise and other human activities
- earthquakes and volcanic activity
- ocean monitoring and natural hazards



Hydroacoustic monitoring equipment prepared for deployment. Source: CTBTO

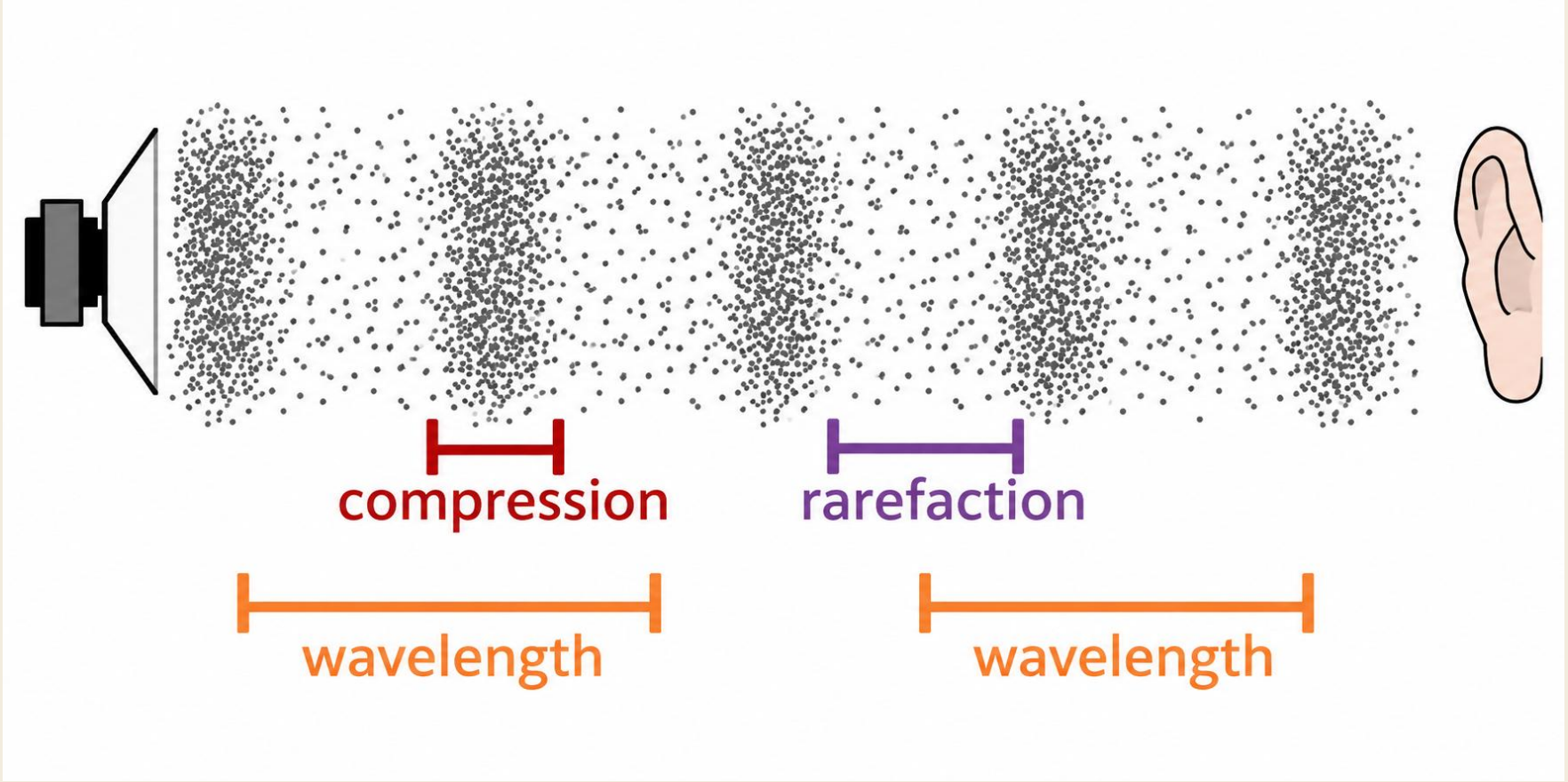
Passive acoustics: Records sounds already present in the environment without transmitting sound.
Active acoustics: Uses deliberately transmitted sound signals, often analysing their echoes or their propagation through water.

What is sound?

Sound is a mechanical wave that propagates through a medium. In water, sound produces both pressure fluctuations and associated particle motion.

As the wave passes:

- water particles oscillate around their equilibrium position
- alternating regions of compression and rarefaction are produced
- energy is transferred through the water



Schematic representation of a longitudinal sound wave. Although air is illustrated, the same principle of compression and rarefaction applies to sound propagating through water.

A hydrophone measures the pressure component of the sound field.

Describing a sound wave

Amplitude
Maximum variation of the signal from its equilibrium value.

Period, T [s]
Time required for one complete oscillation.

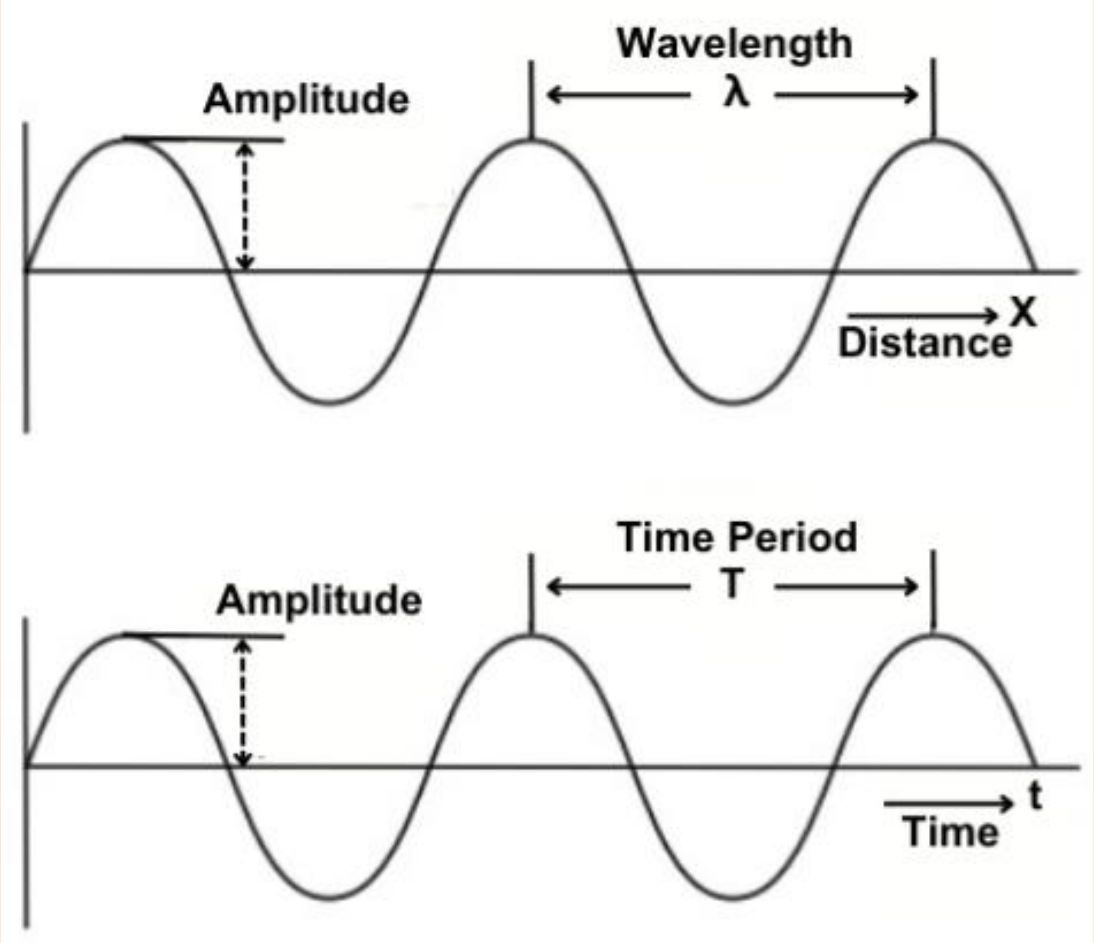
Frequency, f [Hz]
Number of oscillations per second.

$$f = \frac{1}{T}$$

Wavelength, λ [m]
Distance travelled by the wave during one complete oscillation.

$$\lambda = \frac{c}{f}$$

where c [m/s] is the speed of sound.



Quick check
A sound has a period of 0.02 s.
What is its frequency?

Describing a sound wave

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Maximum variation of the signal from its equilibrium value.

Period, T
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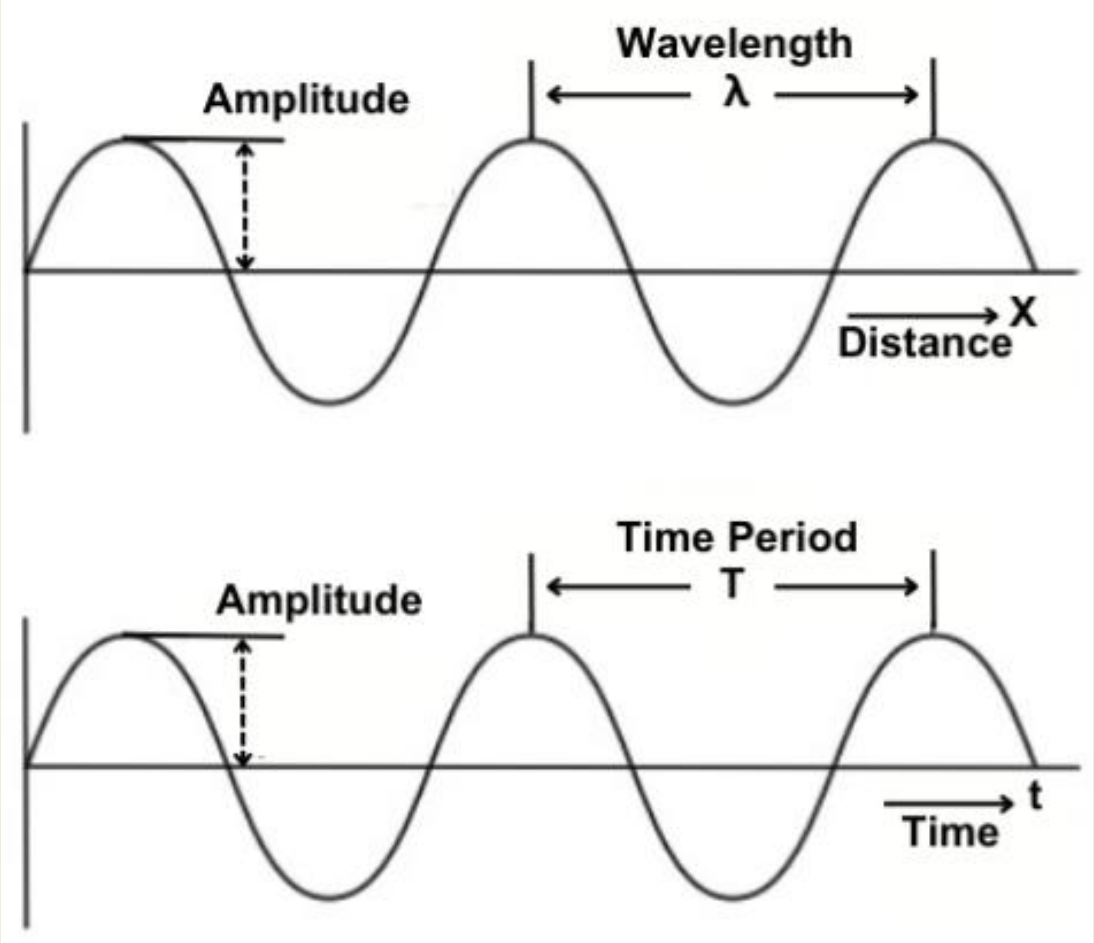
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Distance travelled by the wave during one complete oscillation.

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where c is the speed of sound.



Quick check
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What is its frequency?

Answer: 50 Hz

Frequency: what does Hz mean?

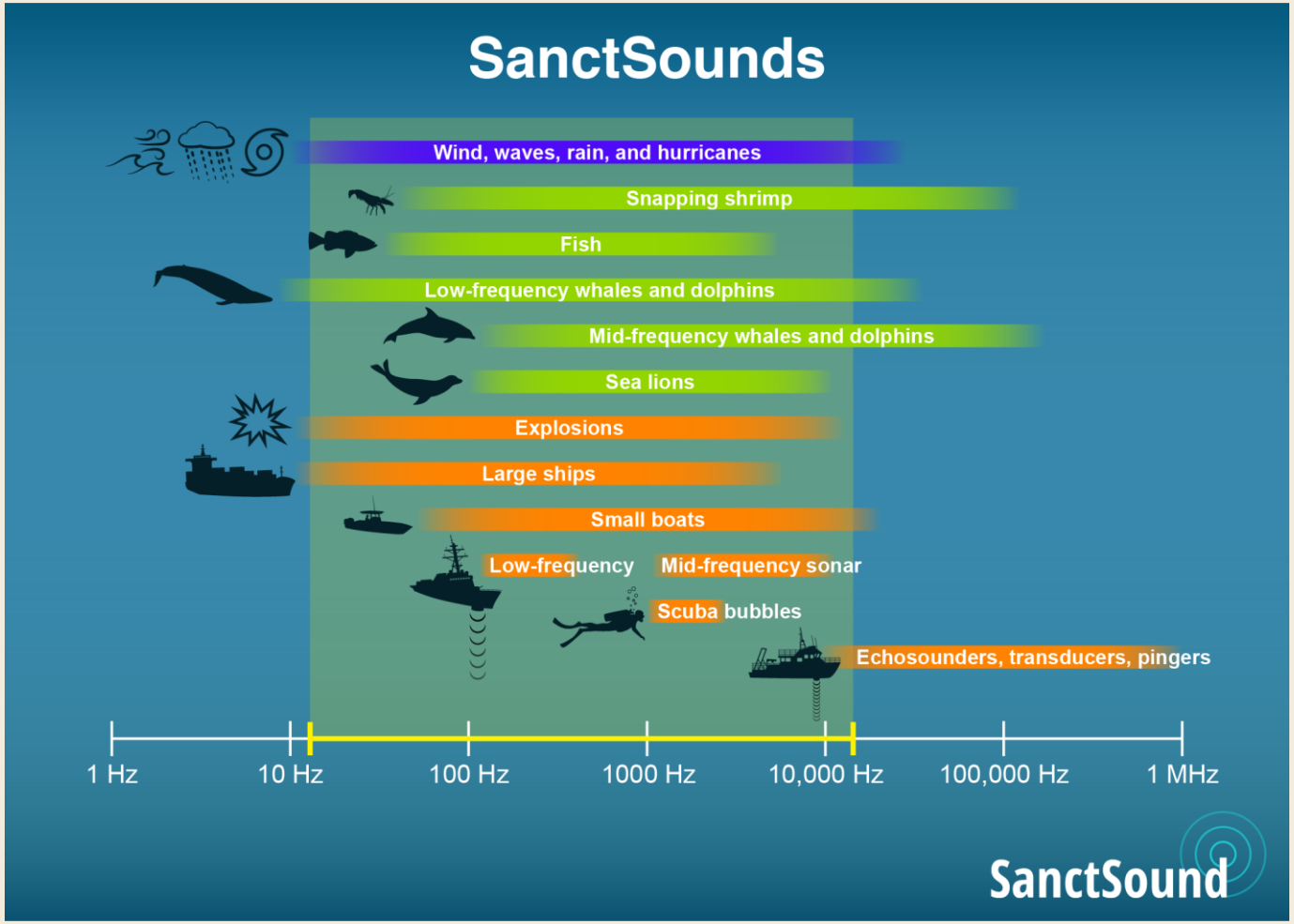
Frequency describes how rapidly a sound oscillates.

1 Hz = one oscillation per second

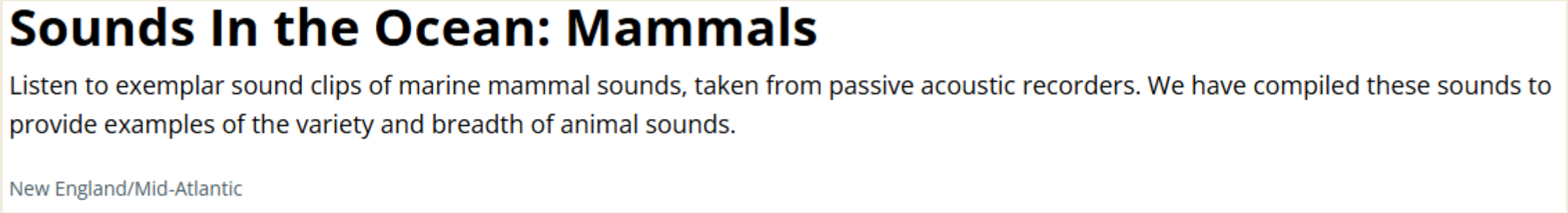
Examples of underwater sounds occur across a very wide frequency range:

- many large-whale calls occur at relatively low frequencies
- vessel noise is often dominated by low frequencies
- dolphin echolocation clicks can contain much higher frequencies

Humans hear only part of the acoustic spectrum present in the ocean!



Source: Aline Design for NOAA https://www.ncei.noaa.gov/news/sanctsound-studying-underwater-world-sound



Listen to different underwater recordings of marine mammals and explore the diversity of sounds they produce.

Can you recognise differences in pitch, rhythm, and sound patterns?

https://www.fisheries.noaa.gov/national/science-data/sounds-ocean-mammals

Amplitude and Sound Pressure Level

Acoustic pressure describes the small pressure fluctuations produced by a sound wave. It is measured in pascals (Pa), often expressed in micropascals (μPa).

Amplitude describes the maximum pressure fluctuation relative to the equilibrium pressure.

Root-mean-square (RMS) pressure describes the effective magnitude of pressure fluctuations over a specified time interval.

Sound Pressure Level (SPL) expresses RMS acoustic pressure on a logarithmic decibel scale:

$$SPL = 20log_{10}(\frac{p_{RMS}}{p_{ref}})$$

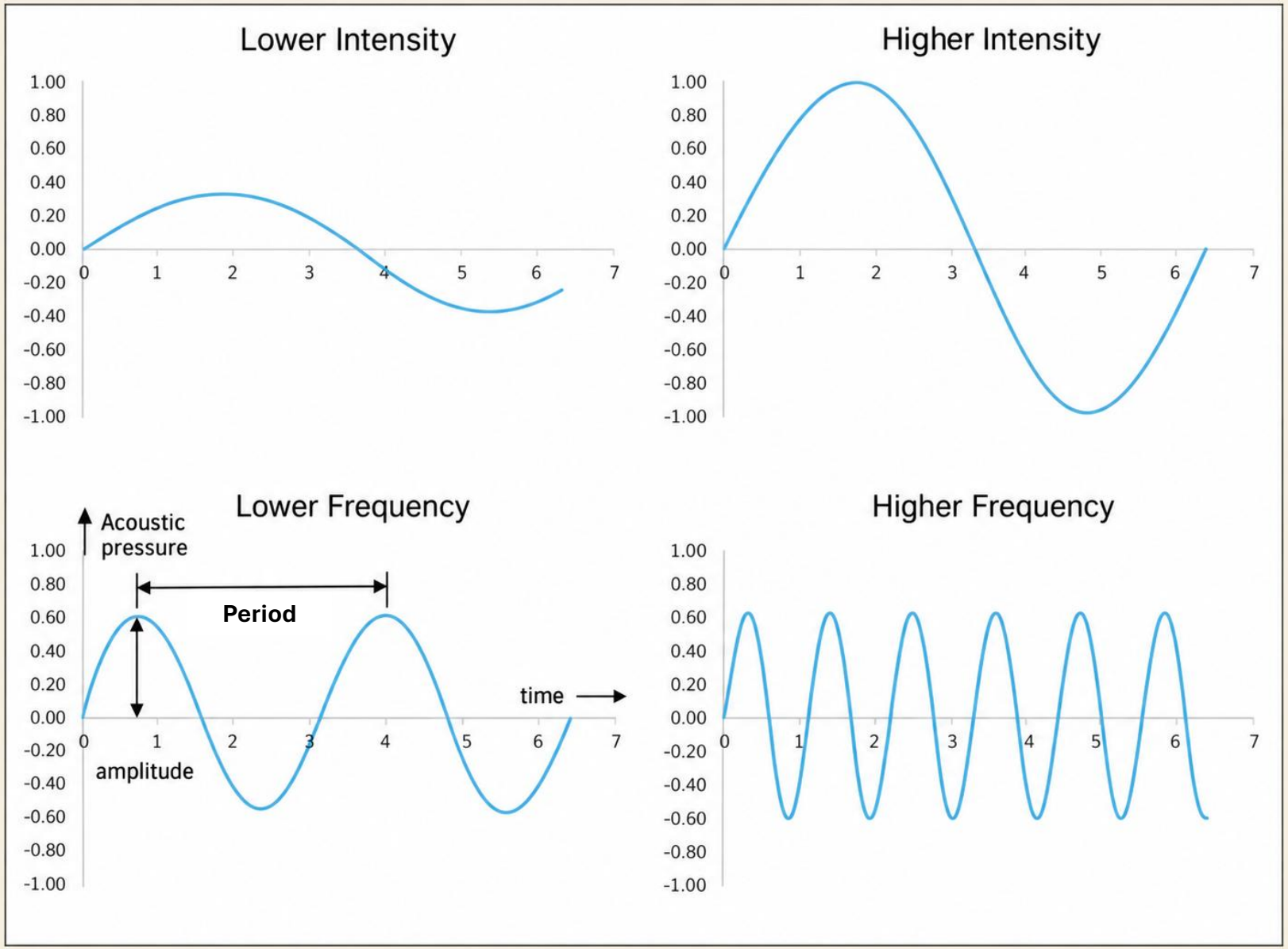
where:

- p_{RMS} : RMS acoustic pressure
- p_{ref} : reference pressure, equal to 1 μPa underwater

Underwater SPL is expressed in **dB re 1 μPa**.
A hydrophone recording must be appropriately calibrated before digital amplitudes can be converted into physical sound pressure levels.

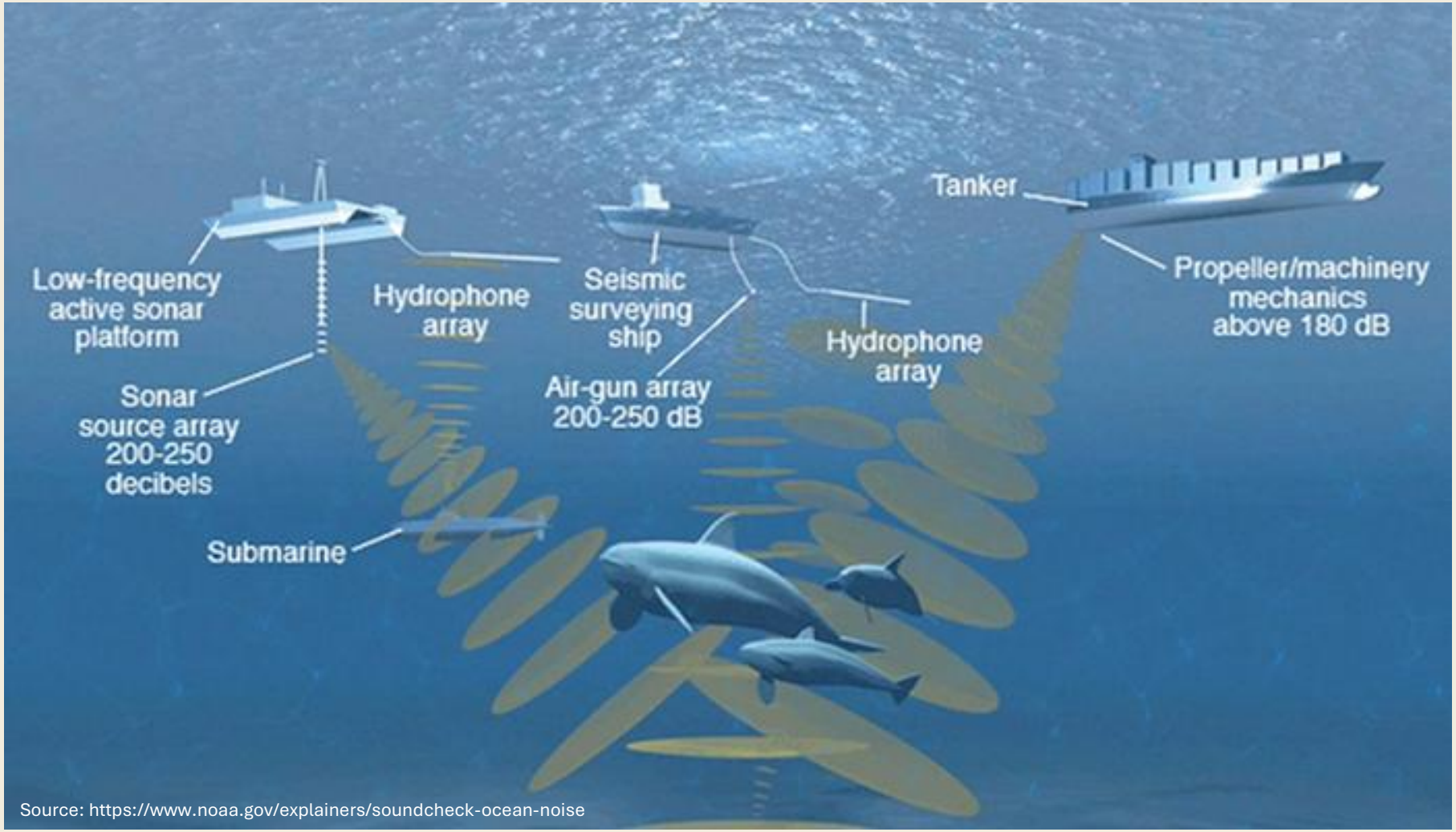
Higher amplitude = larger pressure fluctuations

It does not mean higher frequency



Amplitude and frequency describe different properties of sound

Greater amplitude means larger pressure fluctuations. Higher frequency means more oscillations per second. Increasing one does not necessarily increase the other.



Important: Decibels underwater and in air

Underwater acoustic pressure levels are conventionally referenced to 1 µPa, whereas sound pressure levels in air are commonly referenced to 20 µPa.

Therefore, the same numerical decibel value does not represent the same acoustic pressure in the two environments. Different acoustic properties and measurement conventions must also be considered when making comparisons.

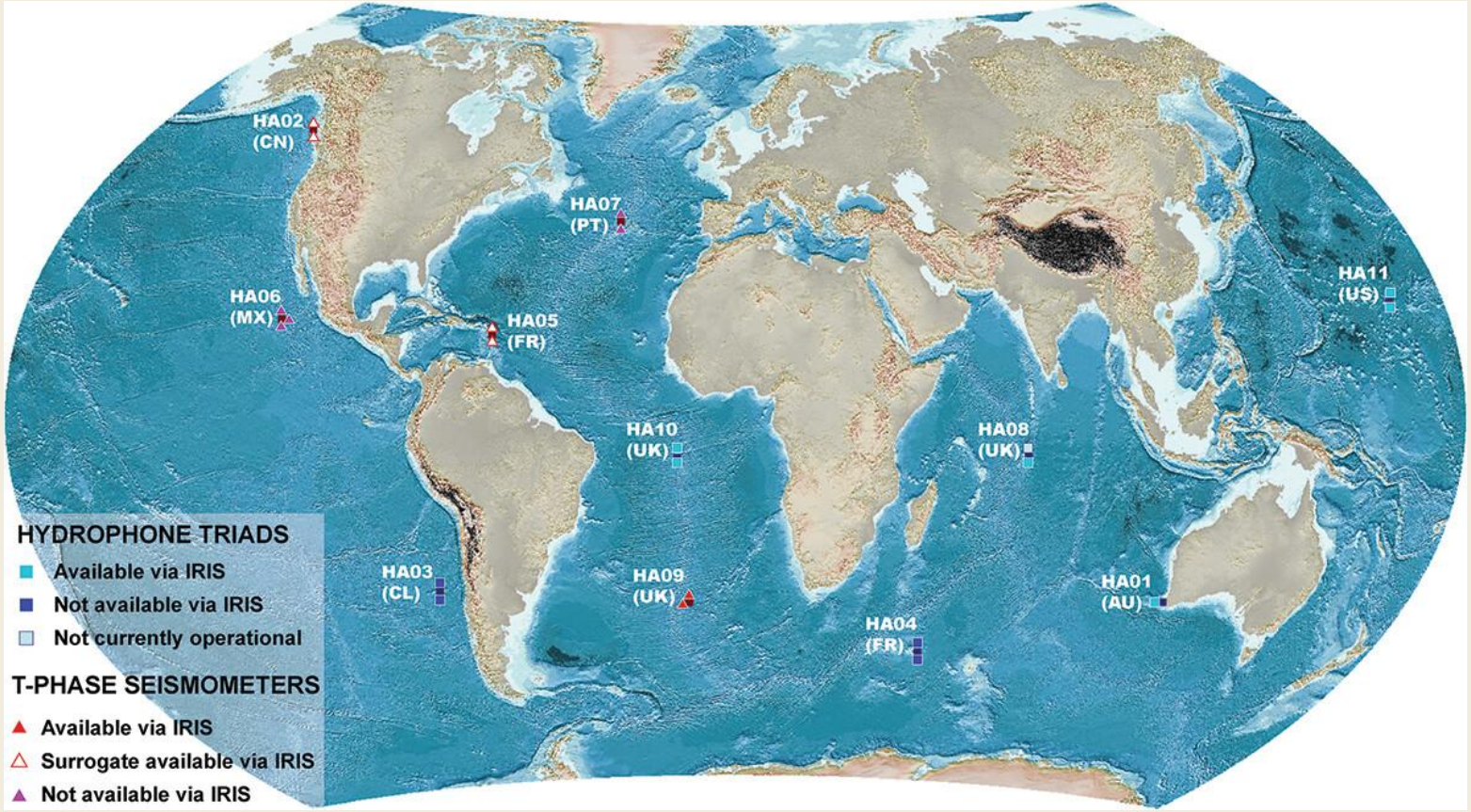
Why is sound so useful underwater?

Light is strongly attenuated in seawater, especially with increasing depth and turbidity.
Sound can travel much greater distances underwater.

This makes acoustics useful for:

- communication
- navigation
- detecting animals or objects
- monitoring human activity
- observing natural events remotely

Hydrophone networks can therefore monitor processes occurring far from the sensor itself.



A world map showing the Comprehensive Nuclear-Test-Ban Treaty Organization International Monitoring System (CTBTO IMS) hydroacoustic component. Source: Gibbons, S. J. (2022). The Hydroacoustic Network of the CTBT International Monitoring System: Access and Exploitation. Journal for Peace and Nuclear Disarmament, 5(2), 452–468. Figure 1. CC BY 4.0.

The CTBTO hydroacoustic network demonstrates how low-frequency underwater sound can be detected across very large distances.
H = hydrophone station
T = seismic T-phase station

Want to know more?

Howe et al. (2019), *Observing the Oceans Acoustically*

frontiers in Marine Science

REVIEW
published: 26 July 2019
doi: 10.3389/fmars.2019.00426

Observing the Oceans Acoustically

Bruce M. Howe[1], Jennifer Miksis-Olds[2], Eric Rehm[3], Hanne Sagen[4], Peter F. Worcester[5] and Georgios Haralabus[6]*

[1] Department of Ocean and Resources Engineering, University of Hawai'i at Mānoa, Honolulu, HI, United States, [2] School of Marine Science and Ocean Engineering, University of New Hampshire, Durham, NH, United States, [3] Takuvik Joint International Laboratory, Université Laval, Quebec City, QC, Canada, [4] Nansen Environmental and Remote Sensing Center, Bergen, Norway, [5] Scripps Institution of Oceanography, University of California, San Diego, La Jolla, CA, United States, [6] Hydro-Acoustics, Engineering and Development, IMS, CTBTO, Vienna, Austria

Acoustics play a central role in humankind's interactions with the ocean and the life within. Passive listening to ocean "soundscapes" informs us about the physical and bio-acoustic environment from earthquakes to communication between fish. Active acoustic probing of the environment informs us about ocean topography, currents and temperature, and abundance and type of marine life vital to fisheries and biodiversity related interests. The two together in a multi-purpose network can lead to discovery and improve understanding of ocean ecosystem health and biodiversity, climate variability and change, and marine hazards and maritime safety. Passive acoustic monitoring (PAM) of sound generated and utilized by marine life as well as other natural (wind, rain, ice, seismics) and anthropogenic (shipping, surveys) sources, has dramatically increased worldwide to enhance understanding of ecological processes. Characterizing ocean soundscapes (the levels and frequency of sound over time and space, and the sources contributing to the sound field), temporal trends in ocean sound at different frequencies, distribution and abundance of marine species that vocalize, and distribution and amount of human activities that generate sound in the sea, all require passive acoustic systems. Acoustic receivers are now routinely acquiring data on a global scale, e.g., Comprehensive Nuclear-Test-Ban Treaty Organization International Monitoring System hydroacoustic arrays, various regional integrated ocean observing systems, and some profiling floats. Judiciously placed low-frequency acoustic sources transmitting to globally distributed PAM and other systems provide: (1) high temporal resolution measurements of large-scale ocean temperature/heat content variability, taking advantage of the inherent integrating nature of acoustic travel-time data using tomography; and (2) acoustic positioning ("underwater GPS") and communication services enabling basin-scale undersea navigation and management of floats, gliders, and AUVs. This will be especially valuable in polar regions with ice cover. Routine deployment of sources during repeat global-scale hydrographic ship surveys would provide high spatial coverage snapshots of ocean temperatures. To fully exploit the PAM systems, precise timing and positioning need to be broadly implemented. Ocean sound is now a mature Global Ocean Observing System (GOOS) "essential ocean variable," which is one crucial step toward providing a fully integrated global multi-purpose ocean acoustic observing system.

Keywords: acoustical oceanography, passive acoustic monitoring, ocean acoustic tomography, acoustic positioning, multi-purpose acoustic networks, marine bio-acoustics, soundscape, essential ocean variable

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What makes sound in the ocean?

Ocean sounds can be grouped into three broad categories.

Biological

- whales
- dolphins
- fish
- snapping shrimp

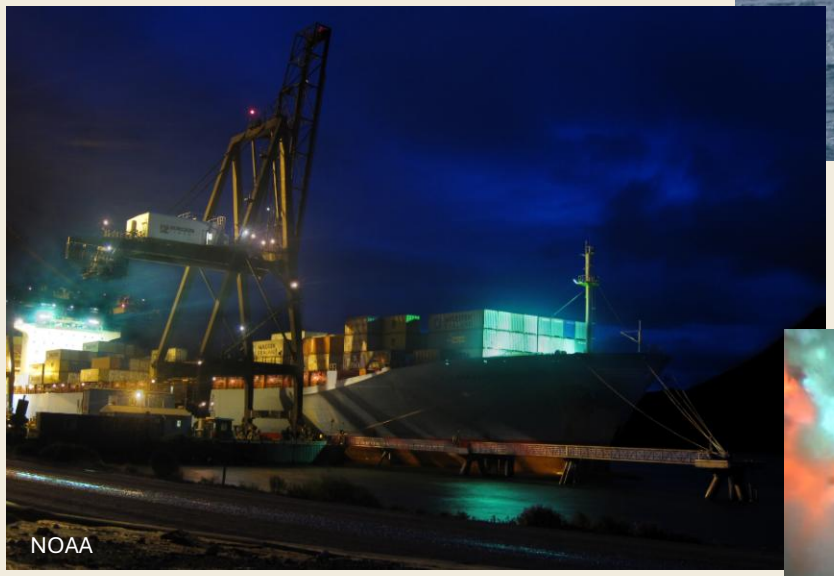
Anthropogenic

- ships
- sonar
- offshore construction
- other machinery

Geophysical and environmental

- earthquakes
- volcanic activity
- waves
- rain
- sea ice







Listen first: can you identify the source?

Listen to each recording before seeing its source.
For each sound, consider:

- What do you think produced it?
- Is the sound continuous or intermittent?
- Does it contain tones, pulses or clicks?
- Does it sound mainly low-frequency or high-frequency?
- What characteristics helped you decide?

- 1st link with recording: https://soundcloud.com/oceannetworkscanada/biggs-orcas-at-folger-passage

- 2nd link with recording: https://soundcloud.com/oceannetworkscanada/sthuqi-sockeye-salmon

- 3rd link with recording: https://soundcloud.com/oceannetworkscanada/large-vessel-travelling-in-the-strait-of-georgia

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A sound as a waveform

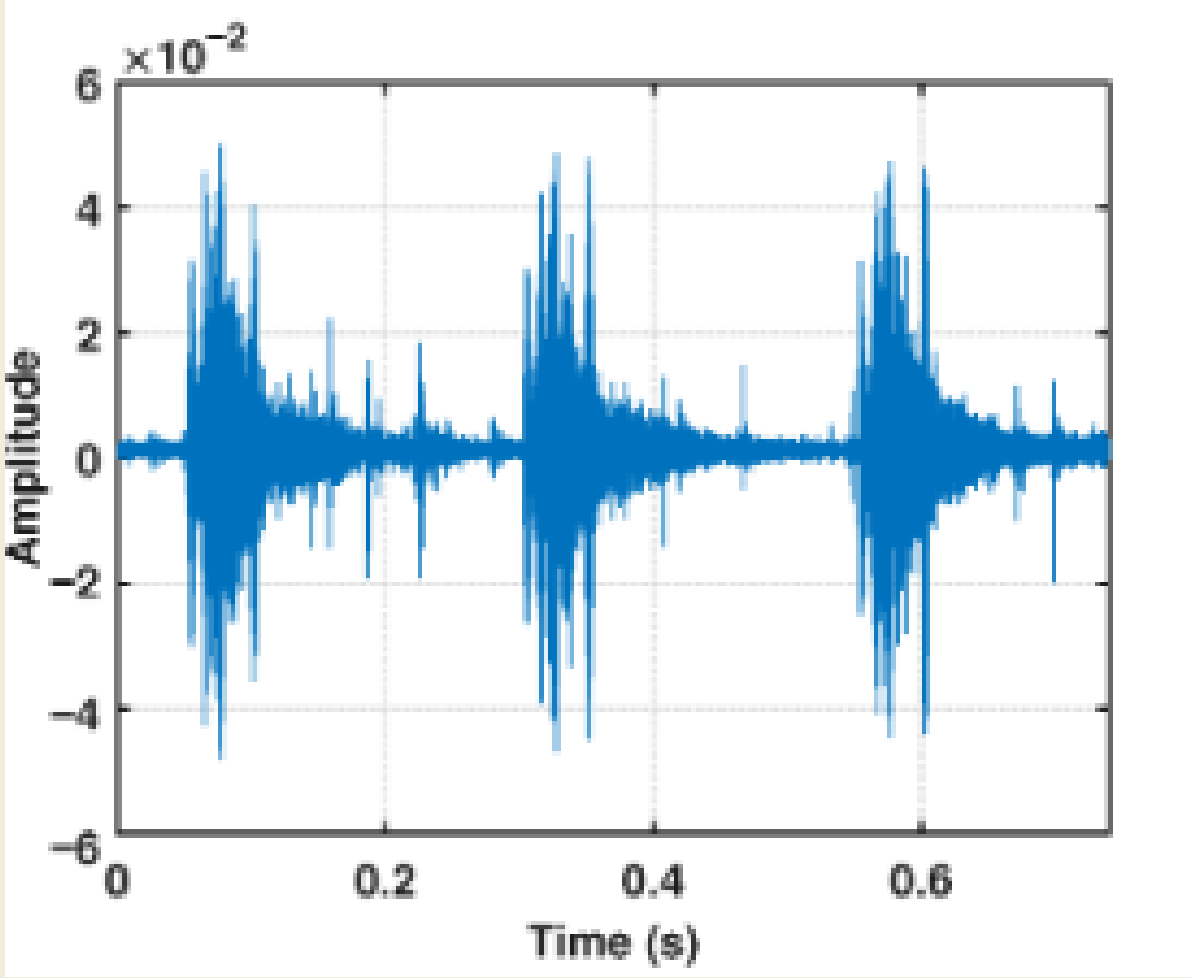
A waveform shows how the acoustic signal changes through time.

- **x-axis:** time
- **y-axis:** pressure or signal amplitude

Waveforms can show:

- when a sound occurs
- how long it lasts
- whether it is continuous or pulsed
- changes in signal amplitude

But different sounds can sometimes produce waveforms that look quite similar.

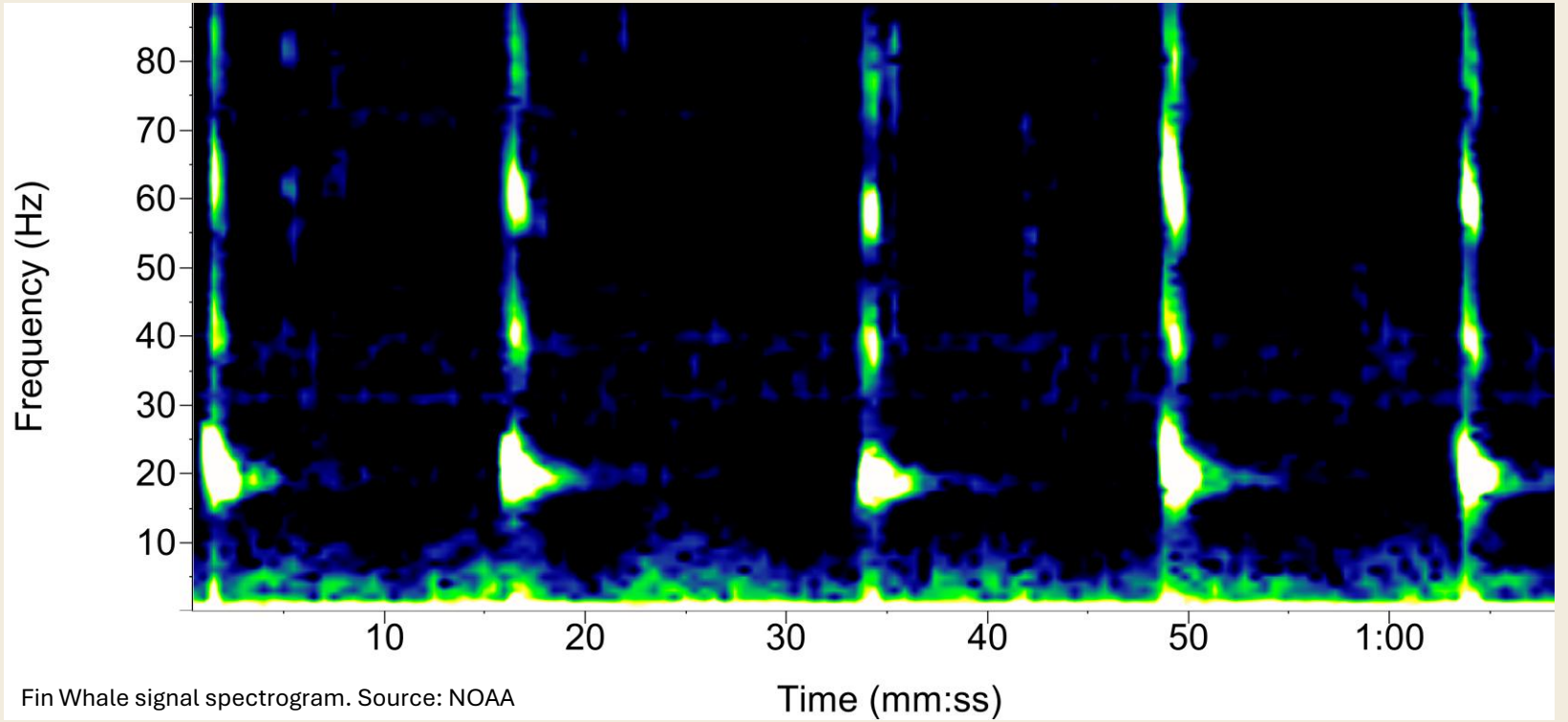


Can you identify the source more easily by listening or by looking at the waveform?

A sound as a spectrogram

A spectrogram shows how the frequency content of a sound changes through time.

- **x-axis:** time
- **y-axis:** frequency
- **colour or intensity:** signal strength



Fin Whale signal spectrogram. Source: NOAA

Spectrograms can reveal features that are difficult to recognise from a waveform alone.

A spectrogram answers: Which frequencies and when?

How to read a spectrogram

- Horizontal axis (time): Shows when sounds occur.
- Vertical axis (frequency): Shows which frequencies are present, measured in hertz (Hz).
- Colours: Dark areas indicate weaker signals, while brighter areas indicate stronger signals (according to the colour scale).

What can we see in this fin whale recording?

- Repeated acoustic pulses occur at approximately regular intervals.
- Strong signal energy appears around 20 Hz, illustrating the low-frequency nature of fin whale calls.
- Each pulse lasts for a relatively short time.
- The spectrogram reveals the timing and frequency structure of these sounds.

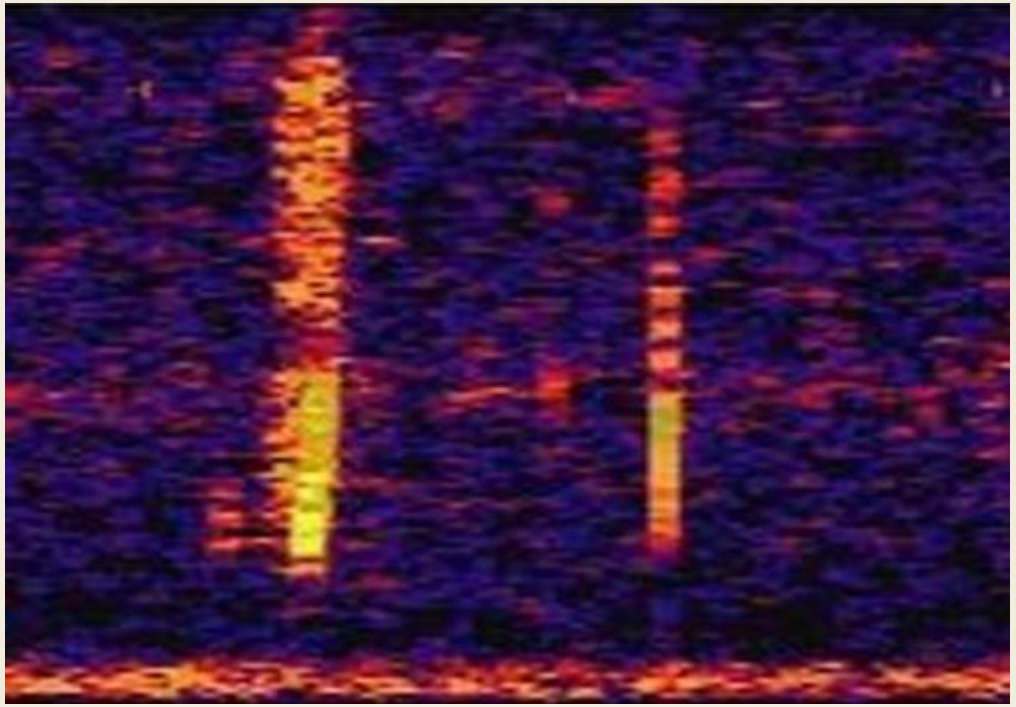
Mystery sound: The Bloop

In 1997, NOAA hydrophones detected an extremely powerful low-frequency underwater sound. The unusual signal became known as “The Bloop.”

Before reading further, listen to the recording
https://www.pmel.noaa.gov/acoustics/sounds/bloop.wav

What do you think produced it?

- a marine animal?
- a ship?
- an earthquake?
- ice?



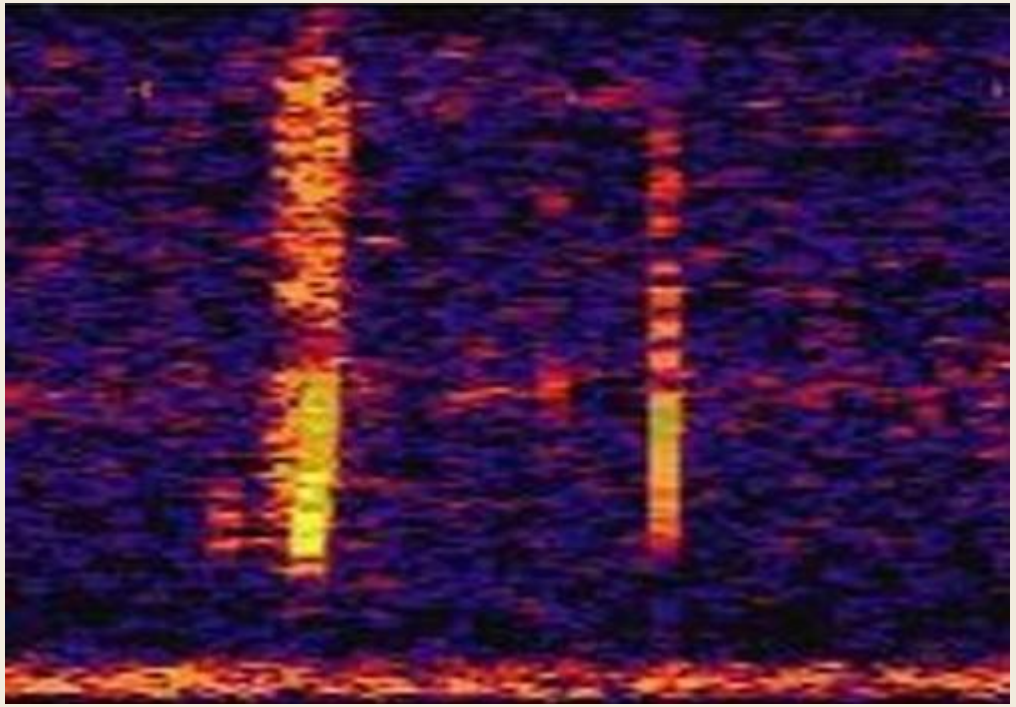
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https://oceanservice.noaa.gov/facts/bloop.html

Explanation

NOAA identified the Bloop as consistent with icequake activity associated with large icebergs cracking and fracturing.
The example illustrates how unfamiliar underwater sounds can initially be difficult to classify, and why additional scientific observations are needed to identify their sources.



Take-home messages

- Underwater sound is a pressure disturbance travelling through water.
- Frequency, amplitude, period and wavelength describe a sound wave.
- Ocean sounds can be biological, anthropogenic or geophysical/environmental.
- Waveform → when does the signal occur?
- Spectrogram → which frequencies occur, and when?
- Hydrophones allow us to observe processes that may be far from the sensor.

Further exploration

Listen

Ocean Networks Canada — real hydrophone recordings of whales, dolphins, Arctic environments and human activity.

NOAA / PMEL — biological, geophysical and environmental underwater sounds.

Watch

Discovery of Sound in the Sea (DOSITS) — short Science of Sound videos and animations.

Read

Howe et al. (2019), *Observing the Oceans Acoustically*, Frontiers in Marine Science, 6:426.

Explore

Ocean Networks Canada Oceans 3.0 — open ocean-observing data and hydrophone recordings.

References

Scientific publications

- Gibbons, S. J. (2022). The Hydroacoustic Network of the CTBT International Monitoring System: Access and Exploitation. *Journal for Peace and Nuclear Disarmament, 5*(2), 452–468. https://doi.org/10.1080/25751654.2022.2129948
- Howe, B. M., et al. (2019). Observing the Oceans Acoustically. *Frontiers in Marine Science, 6*, 426. https://doi.org/10.3389/fmars.2019.00426

Scientific and educational resources

- NOAA NCEI. *SanctSound: Studying the Underwater World of Sound*. https://www.ncei.noaa.gov/news/sanctsound-studying-underwater-world-sound
- NOAA Fisheries. *Sounds in the Ocean: Mammals*. https://www.fisheries.noaa.gov/national/science-data/sounds-ocean-mammals
- NOAA. *Soundcheck: Ocean Noise*. https://www.noaa.gov/explainers/soundcheck-ocean-noise
- NOAA National Ocean Service. *What Is the Bloop?* https://oceanservice.noaa.gov/facts/bloop.html
- Ocean Networks Canada. *Underwater Sound Recordings*. https://www.oceannetworks.ca/multimedia/sounds/
- Discovery of Sound in the Sea (DOSITS). https://dosits.org/

Figure and Media Credits

- **Slide 3:** Hydrophone deployment photograph. CTBTO Public Information (www.ctbto.org).
- **Slide 7:** Underwater sound frequency ranges. Aline Design for NOAA, SanctSound.
- **Slide 8:** Marine mammal sounds resource. NOAA Fisheries.
- **Slide 11:** Underwater noise infographic. NOAA, *Soundcheck: Ocean Noise*.
- **Slide 12:** CTBTO hydroacoustic station map. Gibbons (2022), Figure 1, CC BY 4.0.
- **Slide 13:** Article preview. Howe et al. (2019), *Frontiers in Marine Science*, CC BY 4.0.
- **Slide 14:** Marine wildlife, vessel and underwater volcanic activity photographs. NOAA.
- **Slide 15:** Hydroacoustic recordings. Ocean Networks Canada.
- **Slide 17:** Fin whale spectrogram. NOAA.
- **Slides 18–19:** The Bloop recording and scientific explanation. NOAA National Ocean Service.